Omar Khalil

Junior Machine Learning Engineer

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PROFESSIONAL PROFILE

Machine-learning graduate with strong fundamentals in NLP, embeddings, model evaluation and experimentation. Experienced with PyTorch, Hugging Face and semantic search, with early exposure to RAG and LLM APIs. Technically strong in modeling and data analysis, but less experienced in backend architecture, DevOps and production-grade agent orchestration.

KEY STRENGTHS

- Strong understanding of transformers, embeddings, classification metrics and experiment design.
- Built semantic search and RAG prototypes using FAISS and sentence-transformers.
- Comfortable analyzing model errors quantitatively rather than relying only on qualitative demos.

PROFESSIONAL EXPERIENCE

Machine Learning Intern | VectorSpring Analytics - Amman, Jordan — **Oct 2025 - Jun 2026**

- Trained and evaluated text-classification models for Arabic and English support tickets using scikit-learn and transformer baselines.
- Improved macro F1 from 0.78 to 0.84 by cleaning labels, balancing training data and tuning decision thresholds.
- Built a semantic-search prototype over 25,000 internal text records using sentence-transformers and FAISS.
- Created experiment notebooks and reusable evaluation scripts for precision, recall, F1, confusion matrices and latency.
- Prototyped retrieval-augmented answers with an LLM API and documented common failure modes such as irrelevant retrieval and unsupported claims.

Undergraduate Research Assistant | Applied AI Research Group - University Lab — **Feb 2025 - Sep 2025**

- Prepared Arabic NLP datasets, performed annotation quality checks and compared multilingual transformer models.
- Ran ablation experiments and presented findings to faculty and student researchers.
- Contributed reusable preprocessing and evaluation utilities in Python.

SELECTED PROJECTS

Arabic Semantic Search Engine - *PyTorch, sentence-transformers, FAISS*

Indexed 40,000 Arabic FAQ passages and evaluated retrieval quality with Recall@5 and MRR. Compared multilingual MiniLM and Arabic-specific embedding models.

University Document RAG Prototype - *Python, Hugging Face, FAISS, LLM API*

Built a PDF-to-RAG pipeline with chunking, embeddings and cited answers. Added a simple retrieval-confidence threshold but no advanced agent workflow.

Intent Classifier with Error Analysis - *PyTorch, Transformers, Pandas*

Fine-tuned a multilingual transformer, produced per-class metrics and performed manual error categorization to identify ambiguous labels and data leakage.

TECHNICAL SKILLS

Machine Learning	PyTorch, scikit-learn, Hugging Face Transformers, Pandas, NumPy, model evaluation, feature engineering
NLP / RAG	Embeddings, semantic search, FAISS, chunking, retrieval metrics, prompt design, basic reranking
LLM	OpenAI-compatible APIs, structured prompting, hallucination analysis, basic tool calling
Data	SQL, Jupyter, Matplotlib, data cleaning, experiment tracking basics
Engineering	Python, Git, Linux; FastAPI and Docker at beginner-to-intermediate level
Languages	Arabic - Native | English - Professional working proficiency

EDUCATION

B.Sc. Artificial Intelligence | University of Jordan — **2022 - 2026**

GPA: 3.62/4.00 | Graduation project: Evaluation of Retrieval Methods for Arabic Question Answering

CERTIFICATIONS & LEARNING

- Hugging Face NLP Course - selected modules completed (2025)
- DeepLearning.AI - Machine Learning Specialization (2024)